Taylor Daniels

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ABOUT

Math PhD with expertise in Analytic Number Theory, and growing experience in programming. Currently developing skills in Development, Finance, and Machine Learning—my background is helping me make rapid progress. Passionate about making tools that solve problems.

EDUCATION

PhD, **Mathematics**, Purdue University, West Lafayette, IN — **GPA**: 3.70 — **August 2015 – August 2024**
MS, **Mathematics**, University of Louisville, Louisville, KY — **GPA**: 4.00 — **August 2014 – May 2015**
BS, **Mathematics**, University of Louisville, Louisville, KY — **GPA**: 3.85 — **August 2009 – May 2014**

SKILLS

Programming and Computation: Python, Java-/Typescript, JSON, SQLite, git, XML, XBRL, Xule, HTML, C#, Excel, Mathematica, matplotlib, scipy, bash
ML/DS: Scikit-learn, pandas, data collection and cleaning, text parsing, regression and classification
Math/Physics: Analytic Number Theory, Integer Partitions, Spectral Geometry, Gauge Theory
Certifications: Summer 2026 Data Science Boot Camp – The Erdős Institute
Soft Skills: Excellent communication of technical topics, excellent written and verbal communication, sociable, driven
Languages: English – (Native); **Chinese/Mandarin** – (Proficient); **Spanish** – (Proficient)

EXPERIENCE

Visiting Assistant Professor, Purdue University Mathematics, West Lafayette, IN — **August 2024 – May 2026**
- Lead instructor for 8 sections of 200-level Differential Eqns. and Linear Algebra.

Graduate Teaching Assistant, Purdue University Mathematics, West Lafayette, IN — **August 2015 – August 2024**
- Instructed & graded courses ranging from Calculus up to graduate-level Complex Analysis.
- Awarded Excellent in Teaching by Purdue Mathematics Department in Fall 2020 during the COVID-19 Pandemic.

Instructor, STARTALK Summer Program, Louisville, KY — **Summer 2015**
- Fully planned and taught 4 weeks of 6-hour lessons for a course in Mandarin Chinese for students grade 6-8.

Teaching Assistant, University of Louisville Mathematics, Louisville, KY — **August 2012 – May 2015**
- Tutored and assisted teaching work for undergraduate courses from Algebra to Calculus.

SELECTED PROJECTS

XBRL-based financial dependency graph: — **(Ongoing)**
- Made a self-contained web app that pulls and processes published Treasury yields from US Treasury XML feed.
- *Skills used*: Python, Typescript, SQL, Docker, Financial Mathematics

Bootstrapping Spot Rates from Treasury Yields: [GitHub] — **Fall 2026**
- Made a self-contained web app that pulls and processes published Treasury yields from US Treasury XML feed.
- Made a Python backend to compute long-maturity zero-coupon rates and store results in a local SQL database.
- Made a Typescript web app to plot the results and allow historical data lookup.
- *Skills used*: Python, Typescript, SQL, Docker, Financial Mathematics

Erdős Institute Certificate in Data Science: [GitHub, ExecSummary, Slides, Video] — **Summer 2026**
- Analyzed fair balance and statistical biases in randomly generated Pokémon battles.
- Implemented Javascript tools to reverse-engineer random team generation.
- Developed Python tools to analyze 18,000+ Pokémon Showdown battle logs.
- *Skills used*: Python, Javascript, pandas, Scikit-learn, Data Science, Machine Learning

Doctoral and Professional Research: — **2021 – Present**
- Computed and analyzed long-term asymptotic behaviors of 1000's of novel weighted integer-partition sequences.
- Developed parallel computation algorithms to overcome (sub-)exponential search space; computational speed increased 10-fold.
- *Skills used*: Python, GP/Pari, Sage, remote computation

PUBLICATIONS

Vanishing Coefficients in Products of Quintuple Products, T. Daniels et. al; submitted; arXiv — **2026**
The p*-dissection of a product of quintuple products*, T. Daniels et. al.; submitted; arXiv — **2026**
Vanishings of the Legendre-17 signed partition numbers, T. Daniels; arXiv — **2026**
Legendre-signed partition numbers, T. Daniels, Published in J. Math. Anal. Appl. **542** doi — **2025**
Vanishing coefficients in two q*-series…*, T. Daniels; Published in Res. Number Theory **10**; doi — **2024**
Bounds on the Möbius-signed partition numbers, T. Daniels; Published in Ramanujan J. **65;** doi — **2024**
Biasymptotics of the Möbius- and Liouville-signed partition numbers, T. Daniels, submitted; arXiv — **2023**